

EDA Adventure Works 1

Dimensions Exploration

- Customers come from 6 different countries with the grand majority coming from the United States (followed by Australia)
- There are 4 categories, 36 subcategories and 295 products

From this analysis, I would like to further dig into the following:

1. See what countries we have the most sales in
 2. What bike categories sell the most
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Date Exploration

First Order: 2010-12-29

Last Order: 2014-01-28

Month Range: 37

Oldest customer: 110

Youngest customer: 40

New Customers:

- 2010 - 14 new customers
- 2011 - 2216 new customers
- 2012 - 3225 new customers
- 2013 - 12521 new customers
- 2014 - 506 new customers

Observations:

1. Why did we have so many new customers in 2013? What did we do different?
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Measure Exploration

- Total Sales: \$29,356,250

- Total Quantity Sold: 60,423
 - Average Price: \$486
 - Total Orders: 27,659
 - Total Products: 295
 - Total Customers: 18,484
 - Total Customers Who Have Placed an Order
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Magnitude Analysis

- Total customers by country

	country ↕ 🔍	total_customers ↕ 🔍
1	United States	7482
2	Australia	3591
3	United Kingdom	1913
4	France	1810
5	Germany	1780
6	Canada	1571
7	n/a	337

- Total customers by gender

- Male: 9,341
- Female: 9,128

- Total products by category

	category ↕ 🔍	total_products ↕ 🔍
1	Components	127
2	Bikes	97
3	Clothing	35
4	Accessories	29
5	NULL	7

- What is the average cost in each category

	category ↕ 🔍	avg_cost ↕ 🔍
1	Bikes	949
2	Components	264
3	NULL	28
4	Clothing	24
5	Accessories	13

- Total revenue generated for each category

	category ↕ ⚑	total_revenue ↕ ⚑
1	Bikes	28316272
2	Accessories	700262
3	Clothing	339716

- Distribution of sold items across countries

	country ↕ ⚑	total_sold_items ↕ ⚑
1	United States	20481
2	Australia	13346
3	Canada	7630
4	United Kingdom	6910
5	Germany	5626
6	France	5559
7	n/a	871

Ranking Analysis

- Which five products generate the highest revenue?

	category ↕ ⚑	subcategory ↕ ⚑	product_name ↕ ⚑	total_revenue ↕ ⚑
1	Bikes	Mountain Bikes	Mountain-200 Black- 46	1373454
2	Bikes	Mountain Bikes	Mountain-200 Black- 42	1363128
3	Bikes	Mountain Bikes	Mountain-200 Silver- 38	1339394
4	Bikes	Mountain Bikes	Mountain-200 Silver- 46	1301029
5	Bikes	Mountain Bikes	Mountain-200 Black- 38	1294854

- Which five products generate the lowest revenue?

	category ↕ ⚑	subcategory ↕ ⚑	product_name ↕ ⚑	total_revenue ↕ ⚑
1	Clothing	Socks	Racing Socks- L	2430
2	Clothing	Socks	Racing Socks- M	2682
3	Accessories	Tires and Tubes	Patch Kit/8 Patches	6382
4	Accessories	Cleaners	Bike Wash - Dissolver	7272
5	Accessories	Tires and Tubes	Touring Tire Tube	7440

- Top 10 customers who have generated the highest revenue

	customer_key ↕ ⌵	first_name ↕ ⌵	last_name ↕ ⌵	total_revenue ↕ ⌵
1	1302	Nichole	Nara	13294
2	1133	Kaitlyn	Henderson	13294
3	1309	Margaret	He	13268
4	1132	Randall	Dominguez	13265
5	1301	Adriana	Gonzalez	13242
6	1322	Rosa	Hu	13215
7	1125	Brandi	Gill	13195
8	1308	Brad	She	13172
9	1297	Francisco	Sara	13164
10	434	Maurice	Shan	12914

- What subcategories generated the most revenue?

	category ↕ ⌵	subcategory ↕ ⌵	total_revenue ↕ ⌵
1	Bikes	Road Bikes	14519438
2	Bikes	Mountain Bikes	9952254
3	Bikes	Touring Bikes	3844580
4	Accessories	Tires and Tubes	244634
5	Accessories	Helmets	225435
6	Clothing	Jerseys	173084
7	Clothing	Shorts	71330
8	Accessories	Bottles and Cages	56993
9	Accessories	Fenders	46662
10	Accessories	Hydration Packs	40315
11	Accessories	Bike Stands	39591
12	Accessories	Bike Racks	39360
13	Clothing	Vests	36160
14	Clothing	Gloves	34320
15	Clothing	Caps	19710
16	Accessories	Cleaners	7272
17	Clothing	Socks	5112

EDA Part 2

Change Over Time Analysis

- Quick Summary of KPIs YoY

	order_year	total_sales	total_customers	total_quantity
1	2010	43419	14	14
2	2011	7075088	2216	2216
3	2012	5842231	3255	3397
4	2013	16344878	17427	52807
5	2014	45642	834	1970

Observations:

- Again, here we notice that the year 2013 was significantly higher in statistics than any other year: sales and customers skyrocketed.

Cumulative Analysis

- Running totals for revenue & moving average for price

	order_date	total_sales	running_total_sales	moving_average_price
1	2010-01-01	43419	43419	3101
2	2011-01-01	7075088	7118507	3146
3	2012-01-01	5842231	12960738	2670
4	2013-01-01	16344878	29305616	2080
5	2014-01-01	45642	29351258	1668

Performance Analysis

- Performance for the category classes over the years

	order_year	category	current_sales	avg_sales	diff_avg	avg_change	py_sales	diff_py	py_change
1	2012	Accessories	2146	233303	-231157	Below Avg	NULL	NULL	No Change
2	2013	Accessories	667431	233303	434128	Above Avg	2146	665285	Increase
3	2014	Accessories	30332	233303	-202971	Below Avg	667431	-637099	Decrease
4	2010	Bikes	43419	7077914	-7034495	Below Avg	NULL	NULL	No Change
5	2011	Bikes	7075088	7077914	-2826	Below Avg	43419	7031669	Increase
6	2012	Bikes	5839443	7077914	-1238471	Below Avg	7075088	-1235645	Decrease
7	2013	Bikes	15353707	7077914	8275793	Above Avg	5839443	9514264	Increase
8	2012	Clothing	642	113230	-112588	Below Avg	NULL	NULL	No Change
9	2013	Clothing	323740	113230	210510	Above Avg	642	323098	Increase
10	2014	Clothing	15310	113230	-97920	Below Avg	323740	-308430	Decrease

Observations:

- We can see that in 2013, all of the categories performed above average and that because this year the sales were so much more, this off-set the average sales and thus all other "avg_change" were below.

Part to Whole Analysis

- Comparing the revenue contribution to overall revenue per category

	category	total_sales	overall_sales	percentage_of_total
1	Bikes	28316272	29356250	96.46
2	Accessories	700262	29356250	2.39
3	Clothing	339716	29356250	1.16

Data Segmentation Analysis

- Product cost segmentation

	cost_range	total_products
1	Below 100	110
2	100-500	101
3	500-1000	45
4	Above 1000	39

- Customer segmentation based on status

- VIP: Customers with at least 12 months of history and spending more than €5,000.
- Regular: Customers with at least 12 months of history but spending €5,000 or less.
- New: Customers with a lifespan less than 12 months.

	customer_segment	total_customers
1	New	14631
2	Regular	2198
3	VIP	1655

Observations:

1. Considering that very little of our revenue comes from accessories and clothing, we might consider cutting the number of products from these categories, as we have way too many.